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Predictors of suicidal ideation, suicide attempt and suicide death among people with major depressive disorder: A systematic review and meta-analysis of cohort studies

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1. Introduction

Suicide is a global public health challenge nowadays. Every 40 s, one person may die because of suicide (Nock et al., 2008a). Suicide-related screening or assessment tools such as Tool for Assessment of Suicide Risk (TASR) which has been widely used in clinical practice, could help professionals effectively identify people at suicide risk, thus contributing to suicide prevention and suicide deaths (Wong, 2018; Large, 2018). The crisis interventions for suicide (e.g. help hotline, psychiatric emergency room and cognitive behavioral therapy, etc.) could reduce suicide deaths temporarily, and these interventions might also reduce the prevalence of suicide ideation and attempt (Nordentoft, 2011; Wong, 2018). However, the causes of suicidal ideation, suicide attempt and suicide death may be different and the etiology can be multifactorial, including biological factors, psychological factors, cognitive factors, and environmental factors (Sudol and Mann, 2017; Coentre et al., 2017; Gournellis et al., 2018).

Many people who had died of suicide have mental disorders, especially major depressive disorder (MDD) which occurred in half to two thirds of suicide cases (Hegerl, 2016). The pooled lifetime prevalence of suicide attempt was 31% (95% CI: 27%−34%) and the pooled 1-year prevalence was 8% (95% CI: 3%−14%) (Dong et al., 2019) in 27,340 individuals with MDD. 1.45 suicide deaths and 2.76 suicide attempts per 1000 patient-years were reported among 6934 MDD individuals even during the long-term antidepressants treatment (Braun et al., 2016). Identifying individuals with MDD who are at high risk of suicide is a public health challenge and has important clinical implications (May et al., 2012; Spijker et al., 2010; Witte et al., 2009; Harris and Barraclough, 1997). In terms of prediction, the predictors of suicidal ideation, suicide attempt and suicidal death among MDD individuals may be different (Maniam et al., 2014; Nock et al., 2008b). For example, suicidal ideation is episodic, and has quick onset and short duration. So far, it is difficult to monitor and intervene suicidal ideation in real time. Factors which could predict suicide attempt or deaths may not be predictive for suicidal ideation (Kleiman and Nock, 2018).

Awareness of risk factors for suicide in MDD is crucial for informing clinical practice and suicide interventions (Hawton et al., 2013; May et al., 2012). A meta-analysis in 2013 including 19 studies showed that male gender, family history of psychiatric disorder, previous attempted suicide, anxiety comorbid and severe depression were predictors for suicide death in people with depression which included not only MDD, but melancholia and other mood disorders (Hawton et al., 2013). Although the correlations between suicide and depression had been discussed in many studies (Braun et al., 2016; Maniam et al., 2014; Nock et al., 2008b; Nordentoft, 2011; Wong, 2018), the predictors of suicide in depressed patients are inconsistent. There was no meta-analysis that illustrate predictors of suicidal ideation, suicide attempt and suicide death among people with MDD, respectively. The aim of this study was to conduct a systematic review and meta-analysis of the predictive factors for suicidal behaviors (suicidal ideation, suicide attempt and

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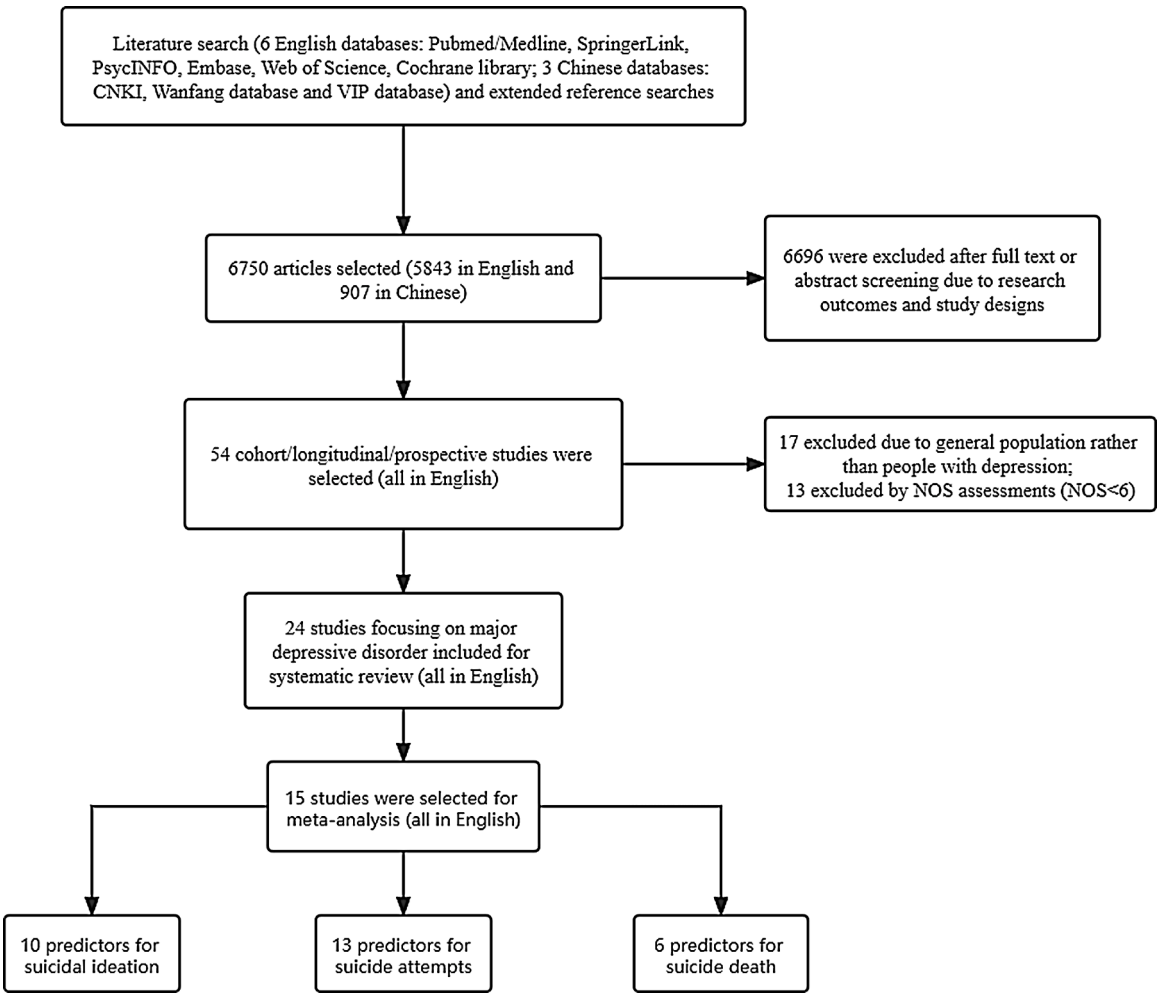


Fig. 1. Flowchart of selection processing

suicide death, respectively) among people with MDD.

2. Methods

2.1. Protocol and registration

The systematic review and meta-analysis was registered in PROSPERO (International Prospective Register of Systematic Reviews) with the online protocol number of CRD42019123324 and was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines (Moher et al., 2009).

2.2. Source and literature searching

Six English electronic databases (PubMed, Embase, Web of Science, SpringerLink, PsycINFO and Cochrane Library) and three Chinese electronic databases (China Knowledge Resource Integrated Database, Wanfang database, and Weipu database) were searched from their inceptions until March 15, 2021. We performed Boolean search technique, e.g., (depression OR depressive disorder* OR major depressive disorder* OR MDD OR depressi* state) AND (suicide OR self-injury OR suicid* thought* OR suicid* ideation* OR suicid* attempt* OR self-injurious OR self-destruction OR self-mutilation OR self-inflicted OR self-slaughter) AND (cohort OR follow-up OR followed-up OR panel OR longitudinal OR prospective). Relevant references were searched as well. The flowchart was shown in Fig. 1.

2.3. Inclusion and exclusion criteria

Inclusion criteria for this study were as follows: (1) Participants had confirmed diagnosis of MDD; (2) Study outcomes included suicidal ideation, suicide attempt and/or suicide death; (3) Studies used cohort or longitudinal design; (4) No restrictions on languages and published periods.

Exclusion criteria for this study were as follows: (1) No reports on suicidal ideation, suicide attempts and suicide death; (2) No reports on MDD; (3) Reports on depressive symptoms; (4) Cross-sectional studies, case-control studies, randomized clinical trials; (5) Did not report or could not calculate hazard ratios (HR), risk ratios or relative risks (RR) and its 95% confidence interval (CI) information; (6) Animal studies or cell studies; (7) Case reports, letters, comments, etc.

2.4. Measurements of major depressive disorder

MDD was diagnosed by clinicians or using fully structured, e.g., Composite International Diagnostic Interview (CIDI), or semi-structured, e.g., Structured Clinical Interview for DSM (SCID) instruments based on Diagnostic and Statistical Manual of Mental Disorders (DSM) or the International Statistical Classification of Diseases (ICD) and Related Health Problems criteria.

2.5. Measurements of suicide related behaviors

Suicidal ideation and suicide attempt were measured by CIDI, ISC

Table 1
Characteristics of 24 studies included (NOS≥6) for systematic review in 954,822 participants with major depressive disorder.

No.	Authors	Published year	Country	Follow-up period	Sample size	MDD patients	Age (years)	Inventory of Depression*	Outcome	Inventory of outcome**	No. of suicide related population	Predictors	NOS score	Meta-analysis included
1	Aaltonen et al.	2019	Finland	24 years	56,826	56,826	≥18	Clinical diagnosed	suicide death	ICD −9	2587	male, depression severity, suicide attempts, depression, alcohol dependence, income, education	6	Yes
2	Alexopoulos et al.	1999	United States	6 months to 7 years (mean= 1.8 years)	354	354	74.17 (mean)	Clinical diagnosed	suicidal ideation	HDRS	155	previous suicide attempts with serious intent, poor social support, impaired instrumental activities of daily living, limited impairment in activities of daily living, severity of depression	6	Yes
3	Bradvik et al.	2008	Sweden	50 years	687	424	0–92	DSM-IV	suicide death	ICD 6–9	28	male, severe depression	7	Yes
4.1	Coryell et al.	2019	United States	16.6 years	838	838	35.9 (mean)	Clinical diagnosed	suicide death	Death registration	23	anxiety symptoms, obsessive-compulsive, somatic, depersonalization, anxiety disorder, obsessive-compulsive disorder, generalized	8	Yes
4.2	Coryell et al.	2019	United States	16.6 years	838	838	35.9 (mean)	Clinical diagnosed	suicide attempts	Self-report	90	anxiety symptoms, obsessive-compulsive, somatic, depersonalization, anxiety disorder, obsessive-compulsive disorder, generalized	8	Yes
5.1	Fergusson et al.	2003	New Zealand	21 years	1063	184	14–21	DSM-IV	suicidal ideation	Self-report	59	family factors, individual factors, school factors	8	Yes
5.2	Fergusson et al.	2003	New Zealand	21 years	1063	184	14–21	DSM-IV	suicide attempts	Self-report	17	family factors, individual factors, school factors	8	Yes
6	Grunebaum et al.	2010	United States	1 year	136	136	39.2 (mean)	DSM-IV	Suicide attempt	SSI	12	Less secure/more avoidant attachment, poor social adjustment	7	No
7.1	Handley et al.	2018	Australia	1, 3 and 5 years	1051	364	53.71 (mean)	CIDI-3.0	suicide attempts	CIDI 3.0	76	depression severity, age, age of depression onset, comorbidities, help-seeking	7	Yes
7.2	Handley et al.	2018	Australia	1, 3 and 5 years	1051	364	53.71 (mean)	CIDI-3.0	suicidal ideation	CIDI 3.0	254	depression severity, age, age of depression onset, comorbidities, help-seeking	7	Yes
8	Hansen et al.	2003	Denmark	10 years	210	210	48.4 (mean)	DSM-III	suicide death	Death registration	15	personality disorder	7	No
9	Holma et al.	2010	Finland	5 years	249	249	39.6 (mean)	DSM–IV	suicide attempts	Beck Scale	36	age, gender, psychotic features, alcohol dependence, marital status, suicide attempt prior to study entry, social support	8	No
10.1	Holma et al.	2020	Finland	5 years	269	269	20–60	SCID-II	suicidal ideation	SSI	15	alcohol abuse/ dependence	6	Yes
10.2	Holma et al.	2020	Finland	5 years	269	269	20–60	SCID-II	suicide attempts	SSI	15	alcohol abuse/dependence	6	Yes
11	Høyer et al.	2004	Denmark	20 years	53,466	22	>15	ICD-8	suicide death	Death registration	1487	gender, diagnosis of alcohol/drug abuse, number of psychiartic admission, admission conditions, duration of illness, time after admission to psychiatric hospital, time after discharge from psychiatric hospital	6	No

(*continued on next page*)

Table 1 (*continued*)

No.	Authors	Published year	Country	Follow-up period	Sample size	MDD patients	Age (years)	Inventory of Depression*	Outcome	Inventory of outcome**	No. of suicide related population	Predictors	NOS score	Meta-analysis included
12	Ilgen et al.	2009	United States	5 years	887,859	885,967	>18	NARDEP	suicide death	ICD-9/ ICD-10	1892	any substance use disorder, those with a substance use disorder (Non-African American), those with a substance use disorder who are not African American (any inpatient treatment in the prior 12-months), Those without a substance use disorder(male),those without a substance use disorder who were male(non-African American)	7	No
13.1	Kovacs et al.	1993	United States	12 -month	183	134	11.2 (mean)	DSM-III	suicidal ideation	ISC items	62	major depressive disorder, comorbid major depressive and conduct/ substance use disorder	7	No
13.2	Kovacs et al.	1993	United States	12 -month	183	134	11.2 (mean)	DSM-III	suicide attempts	ISC items	18	major depressive disorder, comorbid major depressive and conduct/ substance use disorder	7	No
14	Mattisson et al.	2007	Sweden	30–49 years	3563	436	35 (median)	DSM-IV	suicide death	Death registration	12	gender, age at onset, severe impairment, alcohol disorder	6	Yes
15	May et al.	2012	Canada	10 years	49	49	30 (mean)	DSM-IV	suicide attempts	SCID	13	anxiety disorder at baseline, substance abuse at baseline, substance abuse lifetime, personality disorder, hopelessness, social adjustment difficulty, poor maternal relationship	7	Yes
16	Pan et al.	2017	Taiwan	1 year	1710	140	13–14	Adolescent Depression Inventory	suicidal ideation	Self-report	70	BMI category	6	No
17.1	Petteri Sokero et al.	2003	Finland	15 months	269	269	39.6 (mean)	DSM-III	suicidal ideation	SSI	156	sex, age, HAM-D score, HS score, alcohol dependence/abuse, PSSS-R score, SOFAS score	8	Yes
17.2	Petteri Sokero et al.	2003	Finland	15 months	269	269	39.6 (mean)	DSM-III	suicide attempts	SSI	41	sex, age, HAM-D score, HS score, alcohol dependence/abuse, PSSS-R score, SOFAS score	8	Yes
18	Riihimaki et al.	2014	Finland	5 years	134	134	45.3 (mean)	DSM-IV	suicide attempts	SSI	14	gender, age, follow-up time, co-morbid substance use disorder, suicide attempts prior to study entry	6	Yes
19	Schneider et al.	2001	Germany	5 years	278	278	39.8 (mean)	DSM-III-R	suicide death	Self-report	16	hypochondriac delusions and preoccupations, initial insomnia, recurrent depressive episodes	7	No
20	Seo et al.	2014	South Korea	2 years	344	344	46. (mean)	DSM-III-R	suicide attempts	SSI-B	5	Temperament dimension (novelty seeking, harm avoidance, reward dependence, persistence), character dimension (self-directedness, cooperativeness, self-transcendence)	7	No
21	Seo et al.	2015	South Korea	9 years	1003	1003	48.1 (mean)	DSM-IV	suicidal ideation	SSI-B	565	recurrent depression, history of suicide attempt, living alone, single	7	Yes
22.1	Spijker et al.	2010	The Netherlands	2 years	586	248	18–64	CIDI	suicidal ideation	CIDI	97	gender, living without partner, comorbid anxiety, major depressive disorder, previous suicidal ideation, use of professional care, dysthymia	8	Yes
22.2	Spijker et al.	2010	The Netherlands	2 years	586	248	18–64	CIDI	suicide attempts	CIDI	19		8	Yes

(*continued on next page*)

Table 1 (*continued*)

No.	Authors	Published year	Country	Follow-up period	Sample size	MDD patients	Age (years)	Inventory of Depression*	Outcome	Inventory of outcome**	No. of suicide related population	Predictors	NOS score	Meta-analysis included
												living without partner, comorbid anxiety, previous suicidal ideation, previous suicidal acts		
23.1	Wang et al.	2015	United States	3 years	6004	6004	≥18	DSM-IV	suicidal ideation	Self-report	1249	stressful life events	8	Yes
23.2	Wang et al.	2015	United States	3 years	6004	6004	≥18	DSM-IV	suicide attempts	Self-report	169	stressful life events	8	Yes
24	Zivin et al.	2007	United States	5 years	887,859	885,967	>18	NARDEP	suicide death	ICD-9/ ICD-10	1683	gender,race,Hispanic,any substance abuse,age(>65,45–64,18–44), PTSD, previous inpatient stay for psychiatric disorder, service connection, regin, Charison score	7	Yes

Fourth Edition criteria.

DSM-III: The Diagnostic and Statistical Manual of Mental Disorders.

Third Edition criteria.

WMH-CIDI 3.0: World Mental Health Composite International Diagnostic Interview version 3.0.

ICD-8: The International Statistical Classification of Diseases and Related Health Problems 8th Revision.

ICD-9: The International Statistical Classification of Diseases and Related Health Problems 9th Revision.

ICD-10: The International Statistical Classification of Diseases and Related Health Problems 10th Revision.

SCID-II: Structured Clinical Interview for DSM-III-R personality disorders.

NARDEP: the VA's National Registry for Depression.

* DSM-IV: The Diagnostic and Statistical Manual of Mental Disorders

** SSI: Scale for Suicide Ideation; SSI-B: Beck Scale for Suicide Ideation; ISC: Symptom-oriented Interview Schedule for Children; HDRS: Hamilton Depression Rating Scale; CIDI: Composite International Diagnostic Interview; ICD: The International Statistical Classification of Diseases and Related Health Problems; SCID: Structured Clinical Interview for DSM-III-R.

Table 2
The summary of meta-analysis results of predictors for suicide related behaviors in people with major depressive disorder.

Suicide related behaviors	Items	Predictors	References	Statistical index	Combined effects	95% CI Lower level	Upper level	Inverse-squared test *p* value	No. of included studies	Egger's test *p* value *
Suicidal ideation										
	Demographic factors									
		Living alone	Not living alone	RR	1.76	1.24	2.50	0.803	2	–
	History of diseases									
		Recurrent depression	No recurrent depression	RR	3.65	0.72	18.55	0.108	2	–
		Previous suicidal behaviors	No previous suicidal behaviors	RR	3.63	2.57	5.13	0.634	4	–
		Severe depression	No severe depression	RR	1.68	1.17	2.43	<0.001	6	0.247
		Alcohol use related disorders	No alcohol use related disorders	RR	1.37	0.70	2.68	0.070	2	–
	Behaviors									
		Help-seeking	No help-seeking	RR	1.65	1.01	2.69	0.623	3	–
	Environmental factors									
		Negative school factors	No negative school factors	RR	2.13	1.70	2.69	0.743	2	–
		Negative family factors	No negative family factors	RR	1.43	1.28	1.59	<0.001	9	0.002
		Stressful life events	No stressful life events	RR	1.15	1.07	1.24	0.001	16	0.122
		Positive school factors	No positive school factors	RR	0.66	0.36	1.22	<0.001	2	–
Suicide attempts										
	Demographic factors									
		Male	Female	RR	0.99	0.40	2.47	0.267	2	–
	History of diseases									
		Substance misuse	No substance misuse	RR	9.29	3.60	23.99	0.396	3	–
		Previous suicidal behaviors	No previous suicidal behaviors	RR	6.38	2.97	13.72	0.704	3	–
		Anxiety disorders	No anxiety disorders	RR	4.08	1.71	9.72	0.872	2	–
		Alcohol use related disorders	No alcohol use related disorders	RR	3.63	2.03	6.50	0.523	4	–
		Severe depression	No severe depression	RR	2.45	0.88	6.85	<0.001	4	–
		Hopelessness	No hopelessness	RR	1.06	0.97	1.16	0.185	3	–
		Anxiety symptoms	No anxiety symptoms	HR	1.03	1.00	1.05	0.650	8	0.438
		Anxiety disorders	No anxiety disorders	HR	1.03	0.79	1.34	0.729	5	–
	Environmental factors									
		Negative school factors	No negative school factors	RR	3.87	2.61	5.75	0.761	2	–
		Negative family factors	No negative family factors	RR	1.65	1.42	1.93	<0.001	10	0.001
		Stressful life events	No stressful life events	RR	1.34	1.12	1.60	0.001	18	0.225
		Positive school factors	No positive school factors	RR	0.45	0.15	1.33	<0.001	2	–
Suicide death										
	Demographic factors									
		Male	Female	HR	3.27	1.55	6.90	0.057	3	–
	History of diseases									
		Previous suicidal behaviors	No previous suicidal behaviors	HR	2.10	1.92	2.30	0.844	4	–
		Anxiety disorders	No anxiety disorders	HR	1.77	1.11	2.81	0.787	5	–
		Severe depression	No severe depression	HR	1.52	1.16	1.98	<0.001	4	–
		Alcohol or substance abuse	No alcohol or subatance abuse	HR	1.44	1.08	1.91	<0.001	4	–
		Anxiety symptoms	No anxiety symptoms	HR	1.02	0.97	1.07	0.363	8	0.773

* Egger's test was conducted while eight and over studies were included for meta-analysis of each predictor.

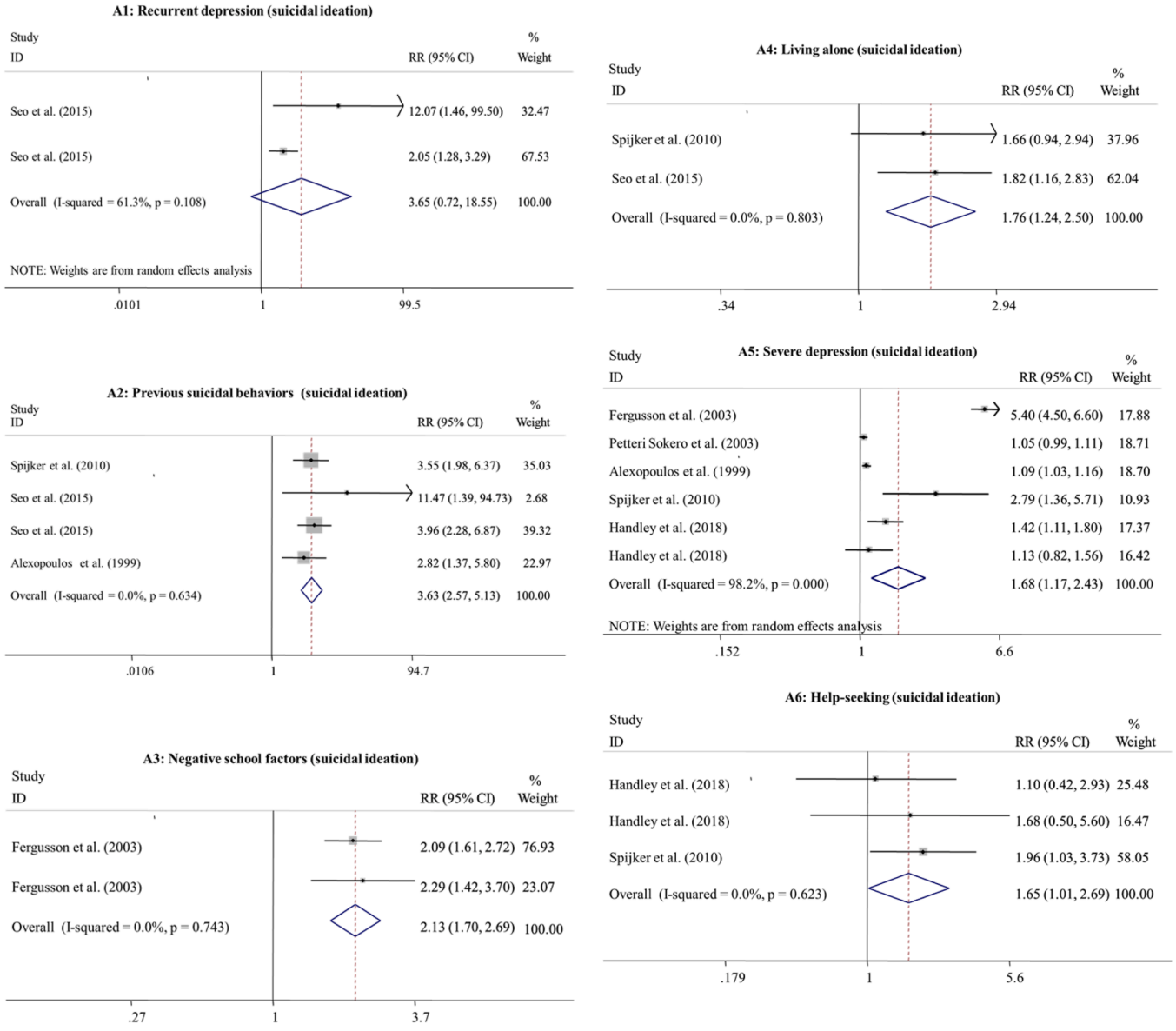
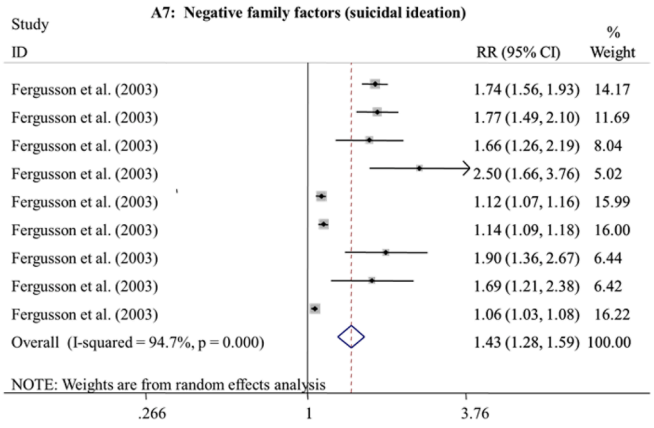
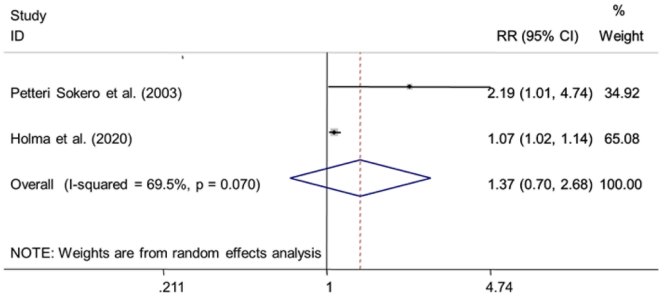
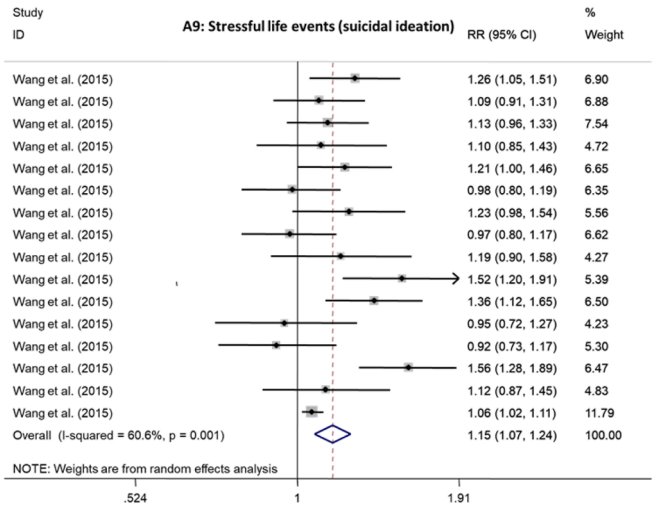
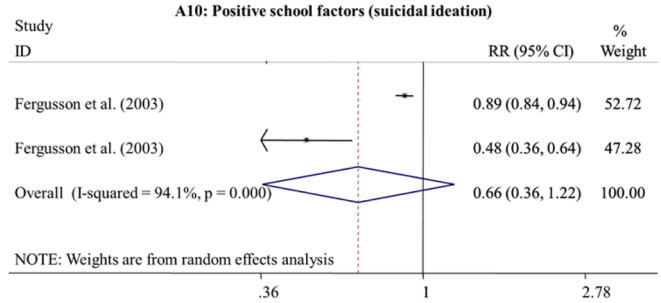


Fig. 2. Pooled estimates with 95% CI for associated factors of suicide related behaviors (suicidal ideation, suicide attempts and suicide, respectively) in people with major depressive disorders (A: Suicidal ideation; A1: Recurrent depression; A2: Previous suicidal behaviors; A3: Negative school factors; A4: Living alone; A5: Severe depression; A6: Help-seeking; A7: Negative family factors; A8: Alcohol use related disorders; A9: Stressful life events; A10: Positive school factors; B: Suicide attempts; B1: Substance misuse; B2: Previous suicidal behaviors; B3: Anxiety disorders; B4: Negative school factors; B5: Alcohol use related disorders; B6: Severe depression; B7: Negative family factors; B8: Stressful life events; B9: Hopelessness; B10: Male; B11: Positive school factors; B12: Anxiety symptoms; B13: Anxiety disorders; C: Suicide; C1: Male;









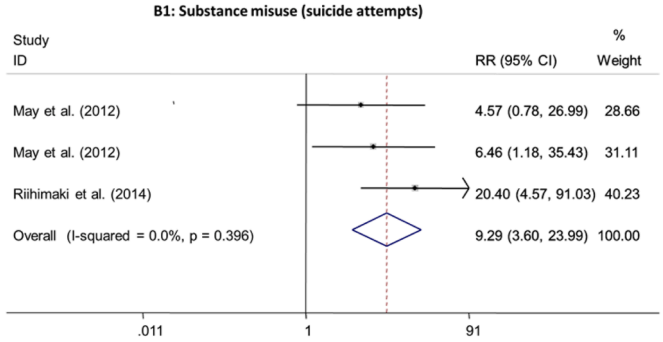


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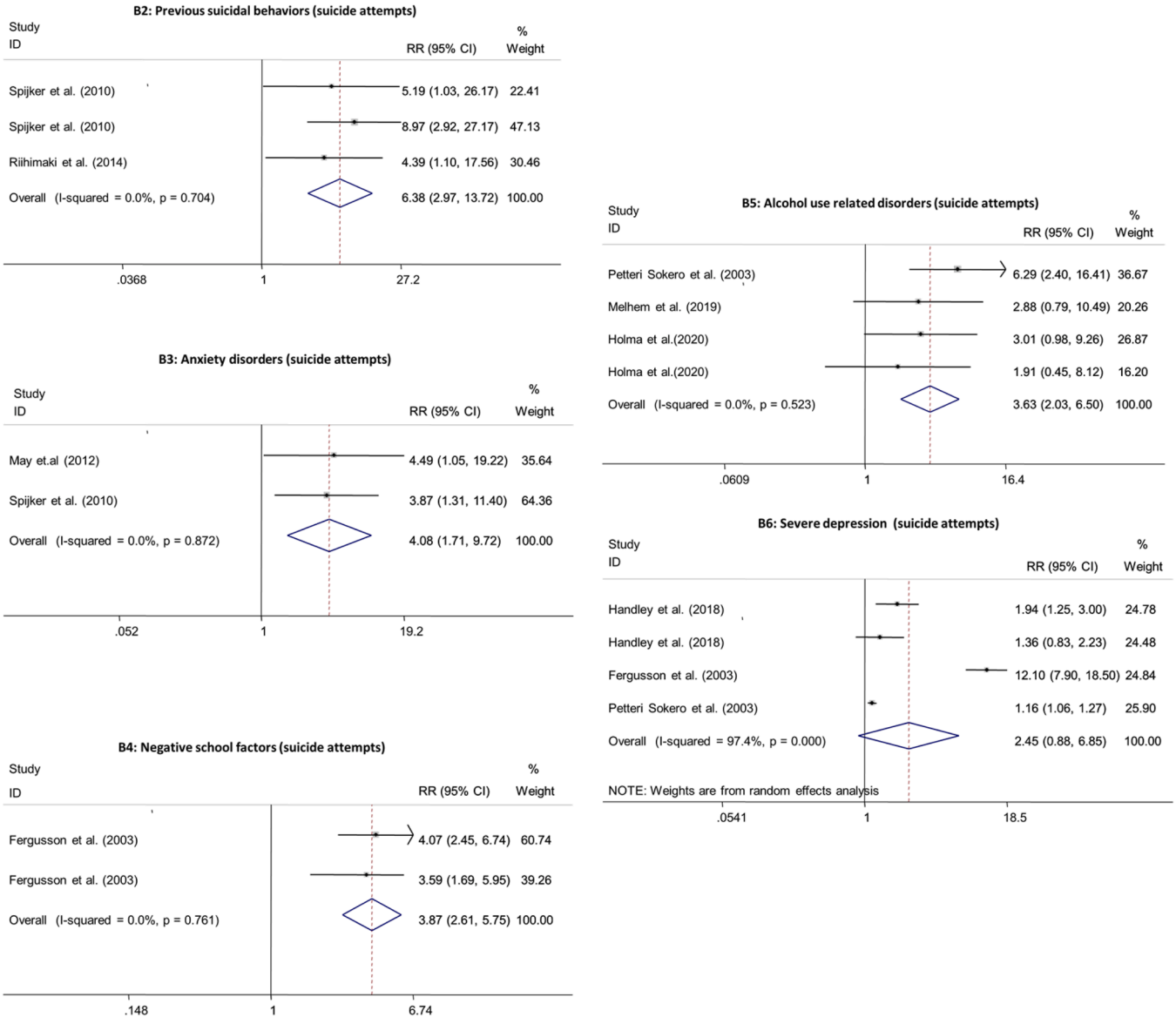


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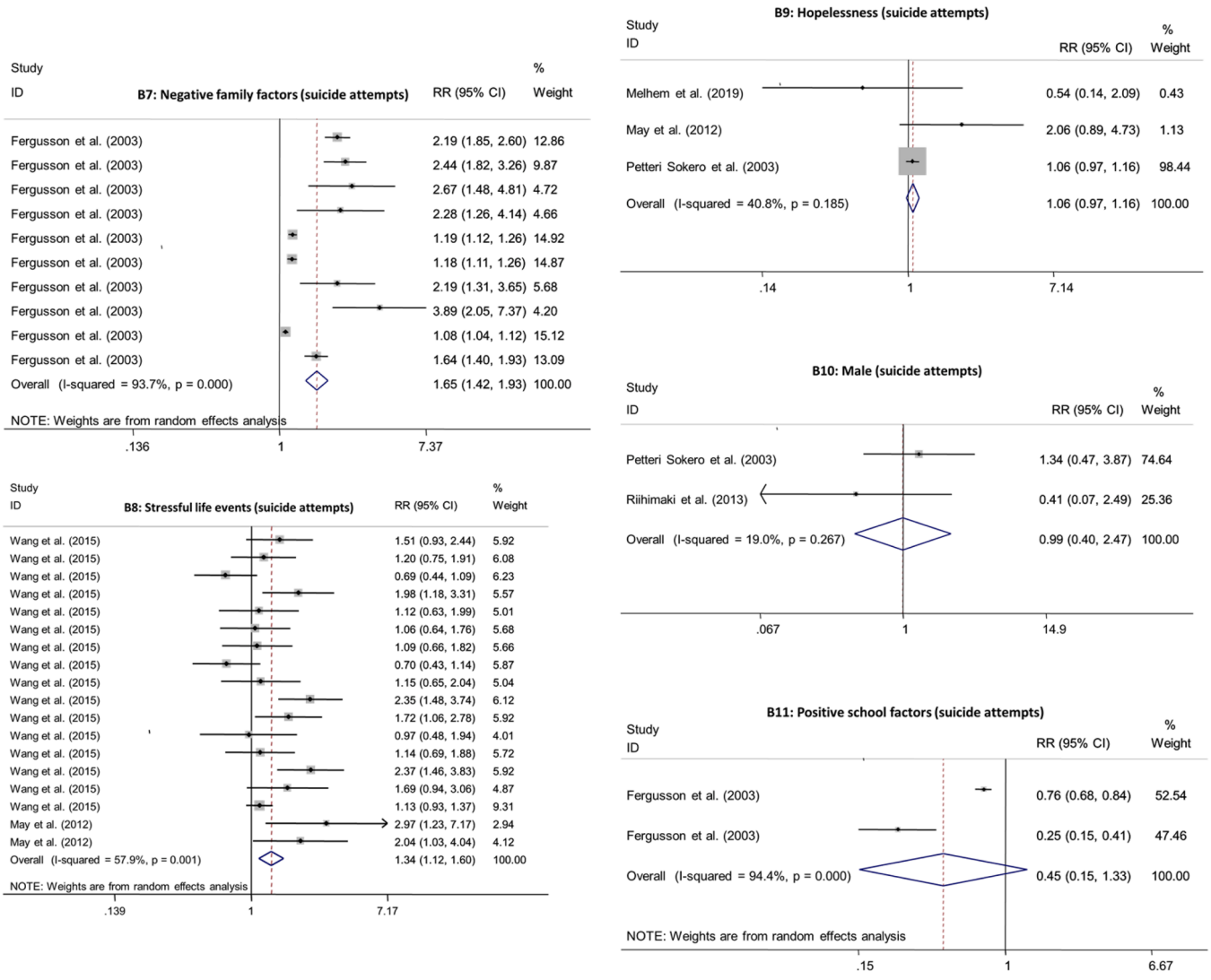
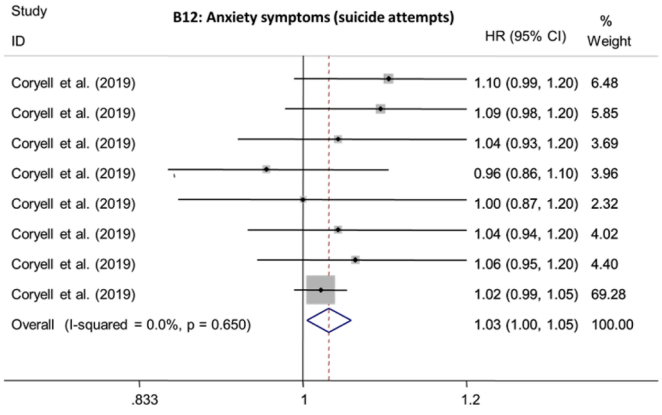
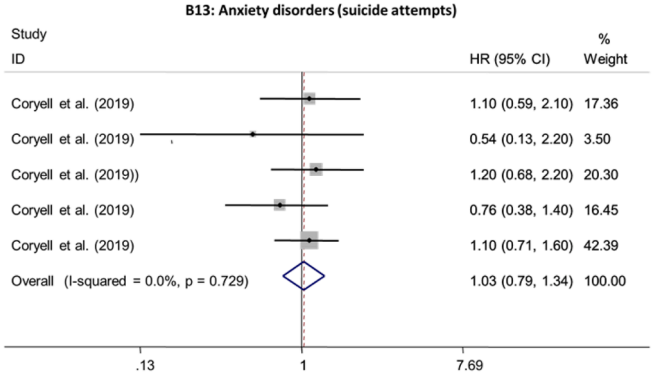
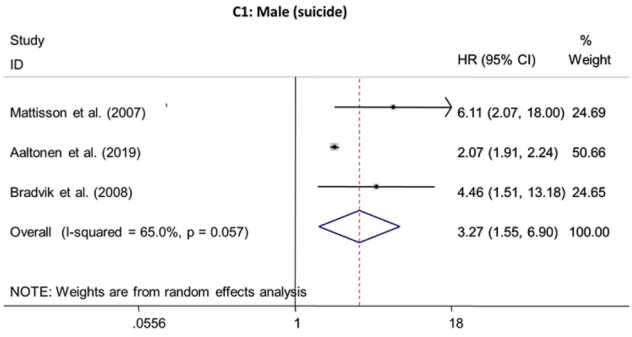
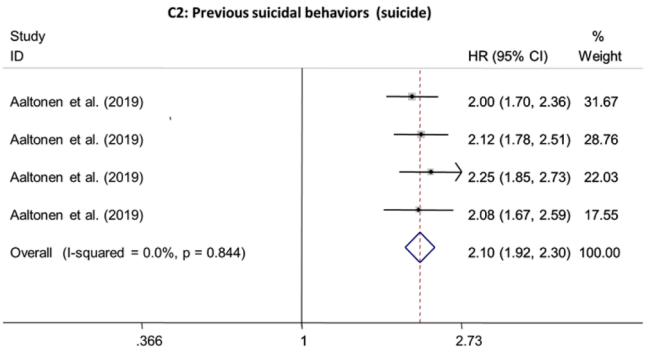


Fig. 2. (*continued*).









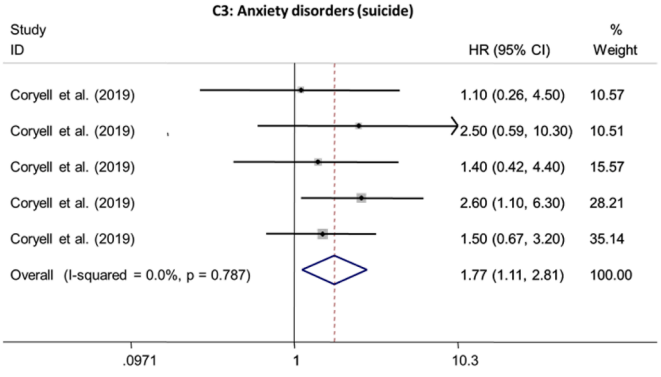


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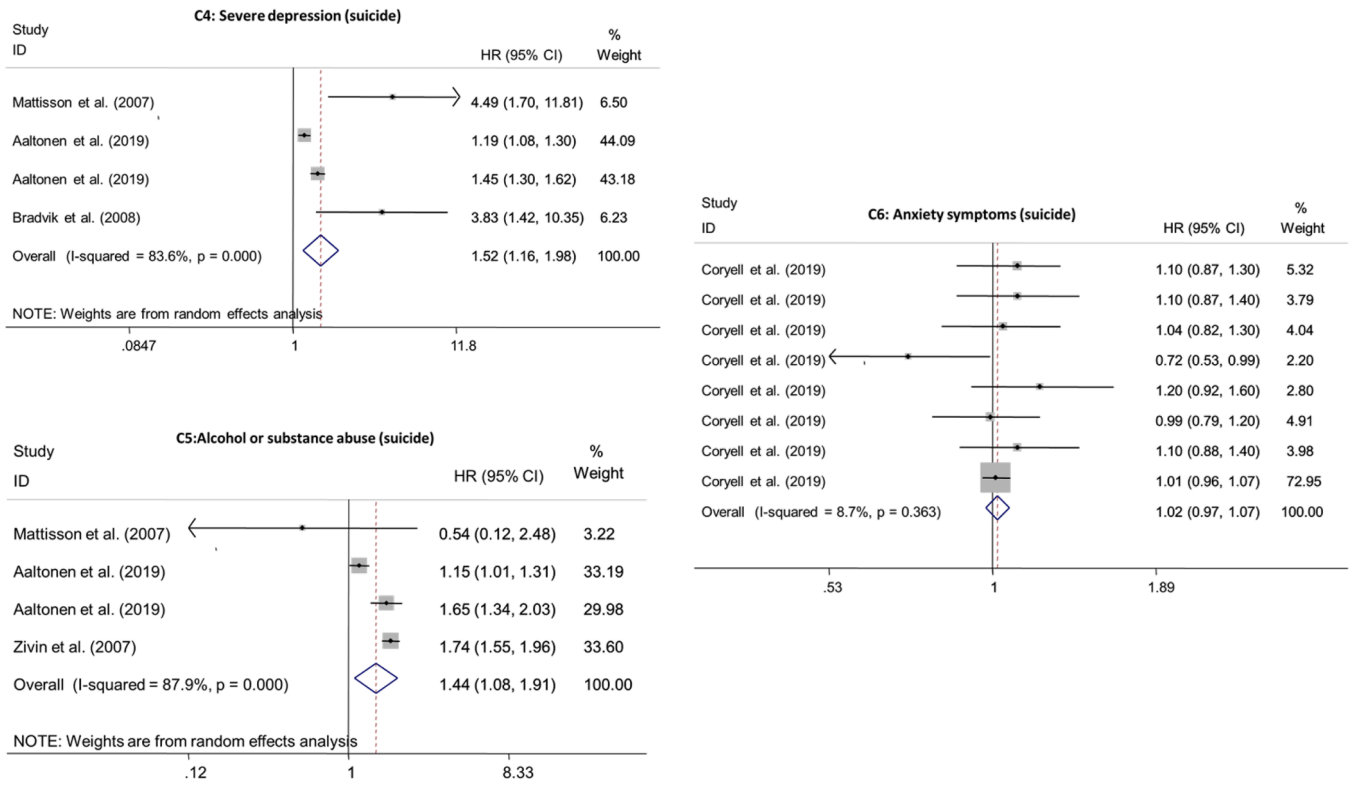


Fig. 2. (*continued*).

(Symptom-oriented Interview Schedule for Children), SSI (the Scale for Suicide Ideation), SSI-B (Beck Scale for Suicide Ideation) or self-reports.

Suicide death was ascertained by ICD codes, death registration or reported by others.

2.6. Quality assessment (NOS criteria)

The quality of selected studies was assessed by Newcastle–Ottawa Scale (NOS) (Stang, 2010). It consisted of 8 items which was divided into 3 dimensions: (1) Selection of the study (4 items); (2) Comparability (1 item); (3) Outcomes for cohort studies (3 items). The total score of NOS is 9. According to the NOS criteria, studies with the scores of 6 and over were regarded as high quality and were included in this study. Articles with the NOS scores less than 6 were recognized as low quality and were excluded.

2.7. Data extraction

After the full text reading and NOS assessment, data were extracted from the qualified reports and evaluated by two independent investigators (XML & DBL). The extracted information included authors, published year, countries, languages, study sample sizes, inventories, and predictors. If there were any disagreements between the two investigators, the third person's (YL) suggestions would be considered.

2.8. Statistical analysis

The extracted data were analyzed using Stata 15.1 (StataCorp LLC). In terms of heterogeneity test, if $I^2 \leq 50\%$, a fixed effect model was used. Subgroup analyses and sensitivity analyses of leave-one-out method were performed to examine the sources of heterogeneity while $I^2 > 50\%$. Random effect model was used if $I^2 > 50\%$. RR (for suicidal ideation, suicide attempt) or HR (for suicide attempt, suicide death) and their 95% CIs were calculated separately for the pooled effects of predictors for suicidal behaviors in people with MDD. Funnel plots of predictors were constructed to examine the potential publication bias.

Egger's test (α=0.05) was conducted to test publication bias if the number of included studies for meta-analysis of predictors were 10 and over.

3. Results

3.1. Characteristics of included studies

Overall, 6750 articles were identified based on the key words and search strategies and 6696 articles were excluded after title and abstract screening due to ineligible research outcomes and study designs. Totally, 54 cohort articles were collected for full text review. 13 articles were excluded due to the low NOS scores (NOS<6) and 17 articles on general population with depressive symptoms were also excluded. Finally, 24 cohort studies covering 954,822 participants with MDD without age and sex restrictions were selected for systematic review and only 15 studies were included for further meta-analysis based on the availability of relevant data. The publication periods ranged from 1993 to 2020. The sample size ranged from 22 to 885,967. The follow-up periods ranged from 1 year to 50 years and the median was 5 years. Seven studies were conducted in United States (29%), followed by five in Finland (21%), two in South Korea (8%), Sweden (8%) and Denmark (8%), respectively. One study was from the Netherlands, Germany, Canada, Taiwan, New Zealand and Australia, respectively. The PRISMA diagram is in Fig. 1. The basic information of all the included 24 cohort studies is presented in Table 1.

3.2. Predictors for suicide related behaviors in people with MDD

Table 2 shows the results of predictive factors associated with suicide related behaviors among people with MDD. The combined effects of predictors for suicidal ideation were: recurrent depression (RR=3.65, 95%CI:0.72–18.55), previous suicidal behaviors (RR=3.63, 95%CI: 2.57–5.13), negative school factors (e.g., truancy, suspension)

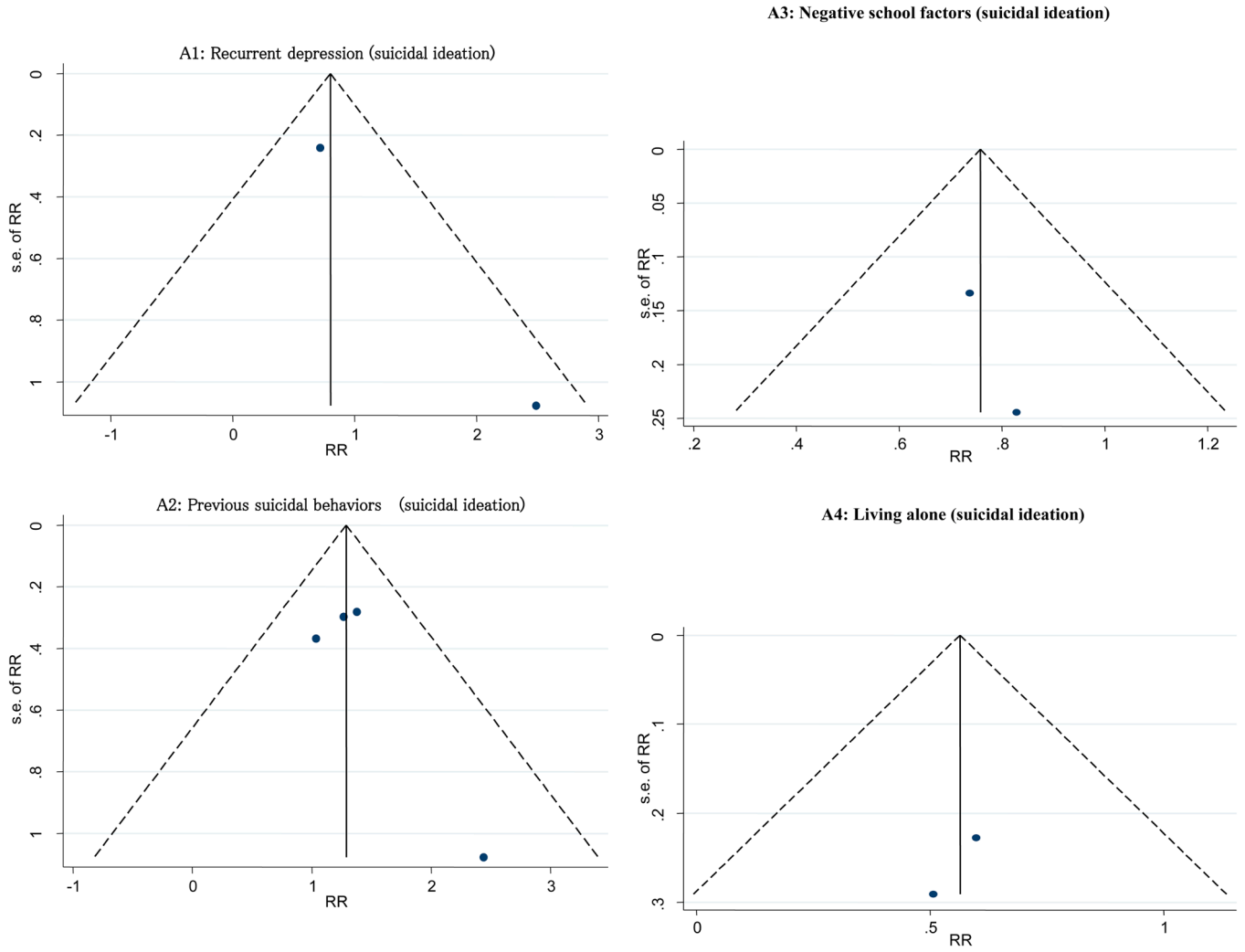


Fig. 3. Funnel plots of associated factors with suicide related behaviors (suicidal ideation, suicide attempts and suicide, respectively) in people with major depressive disorders (A: Suicidal ideation; A1: Recurrent depression; A2: Previous suicidal behaviors; A3: Negative school factors; A4: Living alone; A5: Severe depression; A6: Help-seeking; A7: Negative family factors; A8: Alcohol use related disorders; A9: Stressful life events; A10: Positive school factors; B: Suicide attempts; B1: Substance misuse; B2: Previous suicidal behaviors; B3: Anxiety disorders; B4: Negative school factors; B5: Alcohol use related disorders; B6: Severe depression; B7: Negative family factors; B8: Stressful life events; B9: Hopelessness; B10: Male; B11: Positive school factors; B12: Anxiety symptoms; B13: Anxiety disorders; C: Suicide; C1: Male; C2: Previous suicidal behaviors; C3: Anxiety disorders; C4: Severe depression; C5: Alcohol or substance abuse; C6: Anxiety symptoms).

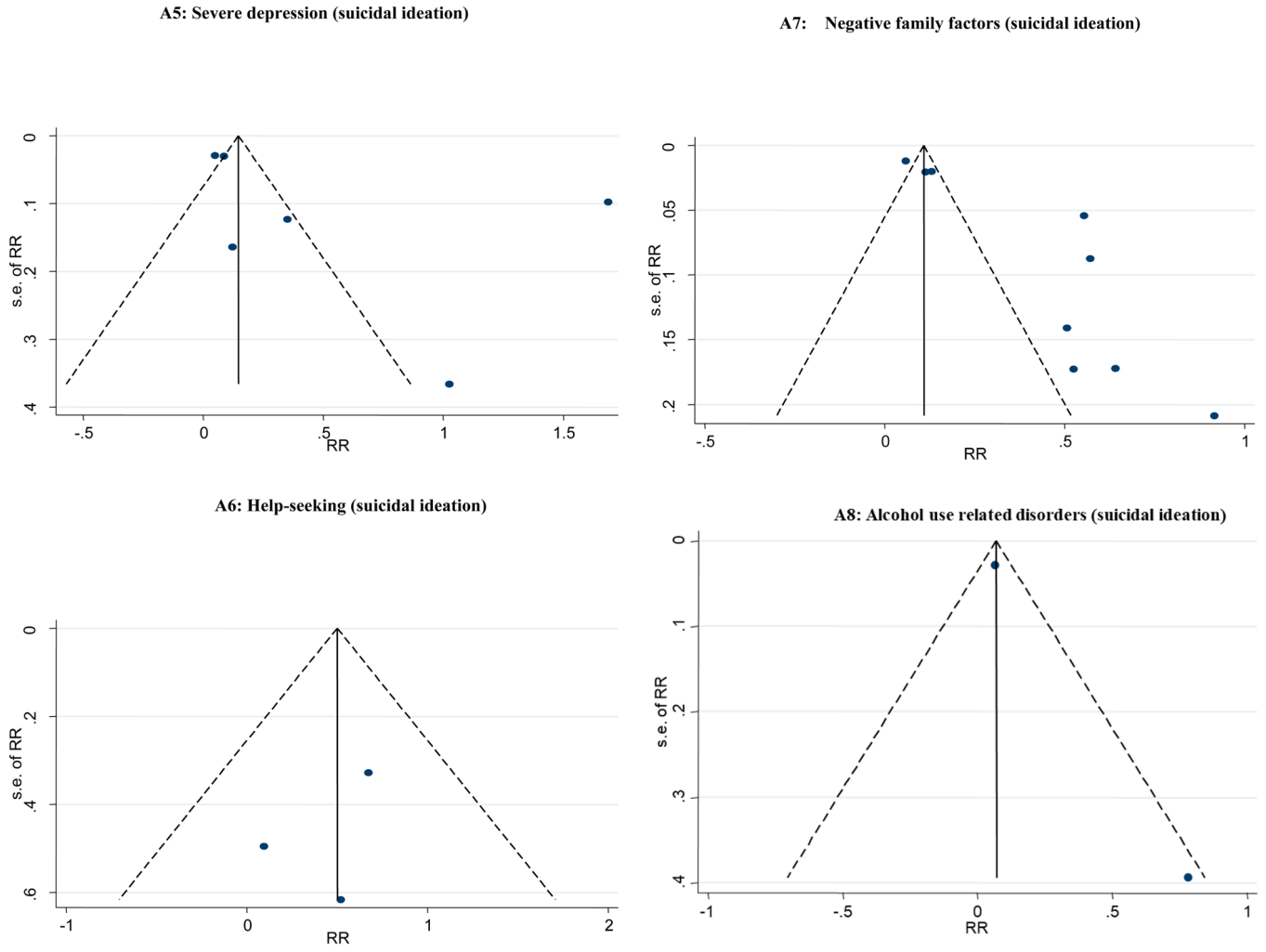


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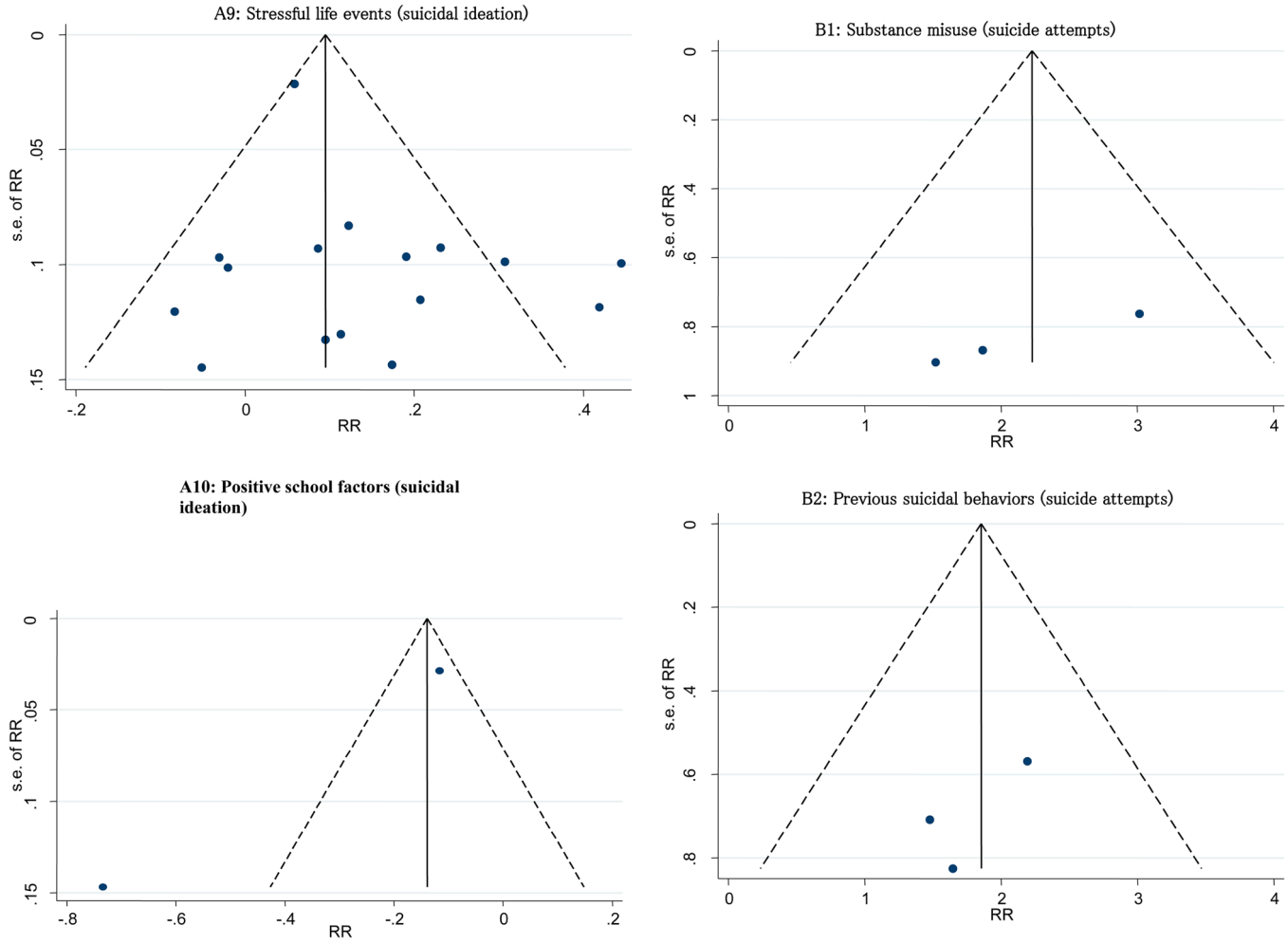


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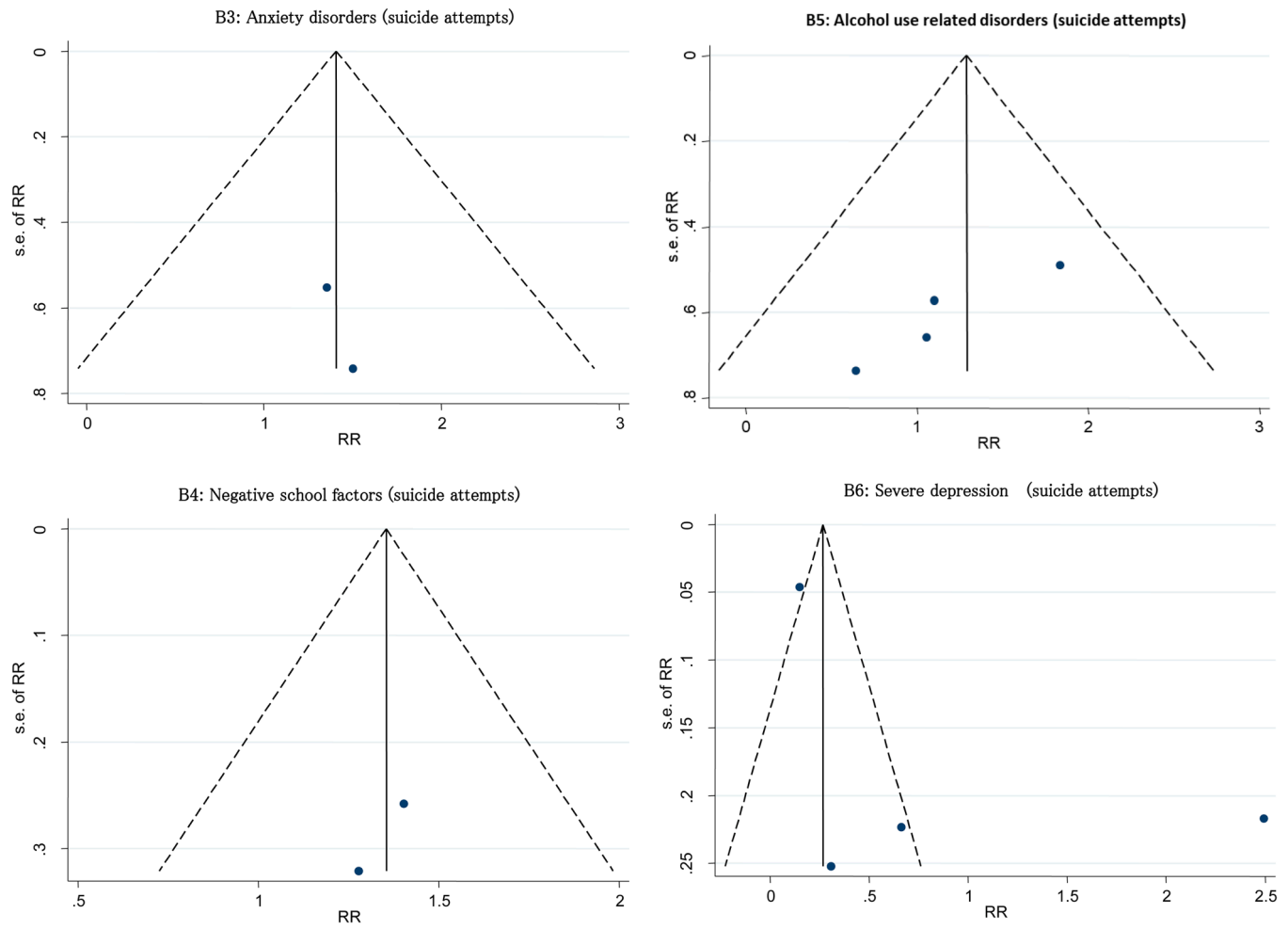


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(RR=2.13, 95%CI: 1.70–2.69), living alone (RR=1.76, 95%CI: 1.24–2.50), severe depression (RR=1.68, 95%CI: 1.17–2.43), help-seeking (e.g., use of professional care) (RR=1.65, 95%CI: 1.01–2.69), negative family factors (e.g., childhood sexual abuse, parents' remarriage, etc.) (RR=1.43, 95%CI: 1.28–1.59), alcohol use related disorders (RR=1.37, 95%CI: 0.70–2.68), stressful life events (e.g., loss, victimization, financial stress, etc.) (RR=1.15, 95%CI: 1.07–1.24) and positive school factors (e.g., school certificate passes, school enjoyment) (RR=0.66, 95%CI: 0.36–1.22). The forest plots were displayed in Fig. 2 (A1∼A10).

The combined effects of the predictors for suicide attempts were: substance misuse (RR=9.29, 95%CI: 3.60–23.99), previous suicidal behaviors (RR=6.38, 95%CI: 2.97–13.72), anxiety disorders (RR=4.08, 95%CI: 1.71–9.72), negative school factors (e.g., truancy, suspension) (RR=3.87, 95%CI: 2.61–5.75), alcohol use related disorders (RR=3.63, 95%CI: 2.03–6.50), severe depression (RR=2.45, 95%CI: 0.88–6.85), negative family factors (e.g., childhood sexual abuse, parents' remarriage, etc.) (RR=1.65, 95%CI: 1.42–1.93), stressful life events (e.g., loss, victimization, financial stress, etc.) (RR=1.34, 95%CI: 1.12–1.60), hopelessness (RR=1.06, 95%CI: 0.97–1.16), male (RR=0.99, 95%CI: 0.40–2.47), positive school factors (e.g., school certificate passes, school enjoyment) (RR=0.45, 95%CI: 0.15–1.33), anxiety symptoms (HR=1.03, 95%CI: 1.00–1.05) and anxiety disorders (HR=1.03, 95%CI: 0.79–1.34). The forest plots were displayed in Fig. 2 (B1∼B13).

The combined effects of the predictors for suicide death were: being male (HR=3.27, 95%CI: 1.55–6.90), previous suicidal behaviors (HR=2.10, 95%CI: 1.92–2.30), anxiety disorders (HR=1.77, 95%CI: 1.11–2.81), severe depression (HR=1.52, 95%CI: 1.16–1.98), alcohol or substance abuse (HR=1.44, 95%CI: 1.08–1.91) and anxiety symptoms (HR=1.02, 95%CI: 0.97–1.07). The forest plots were displayed in Fig. 2 (C1∼C6).

Positive school factors (pooled RR=0.66, 95%CI: 0.36–1.22) appeared to be protective for suicidal ideation, and male sex (pooled RR=0.45, 95%CI: 0.15–1.33) was associated with a lower risk of suicide attempt. However, the associations were not statistically significant.

3.3. Sensitivity analysis and subgroup analysis

For the heterogeneity test results of I^2>50%, we conducted the leave-one-out sensitivity analysis. However, we did not find any study that significantly affected the overall meta-analysis results in each group. Therefore, random effect models were used for the pooled effects calculation. The number of included studies was too small. As such, subgroup analysis was not conducted.

3.4. Funnel plots and Egger's tests for publication bias

Fig. 3 displays the publication bias for each predictive factor. The data showed that there was publication bias associated with recurrent depression and stressful life events for suicidal ideation, negative school factors, positive school factors and negative family factors for suicidal ideation and attempts, previous suicidal behaviors for suicide, anxiety

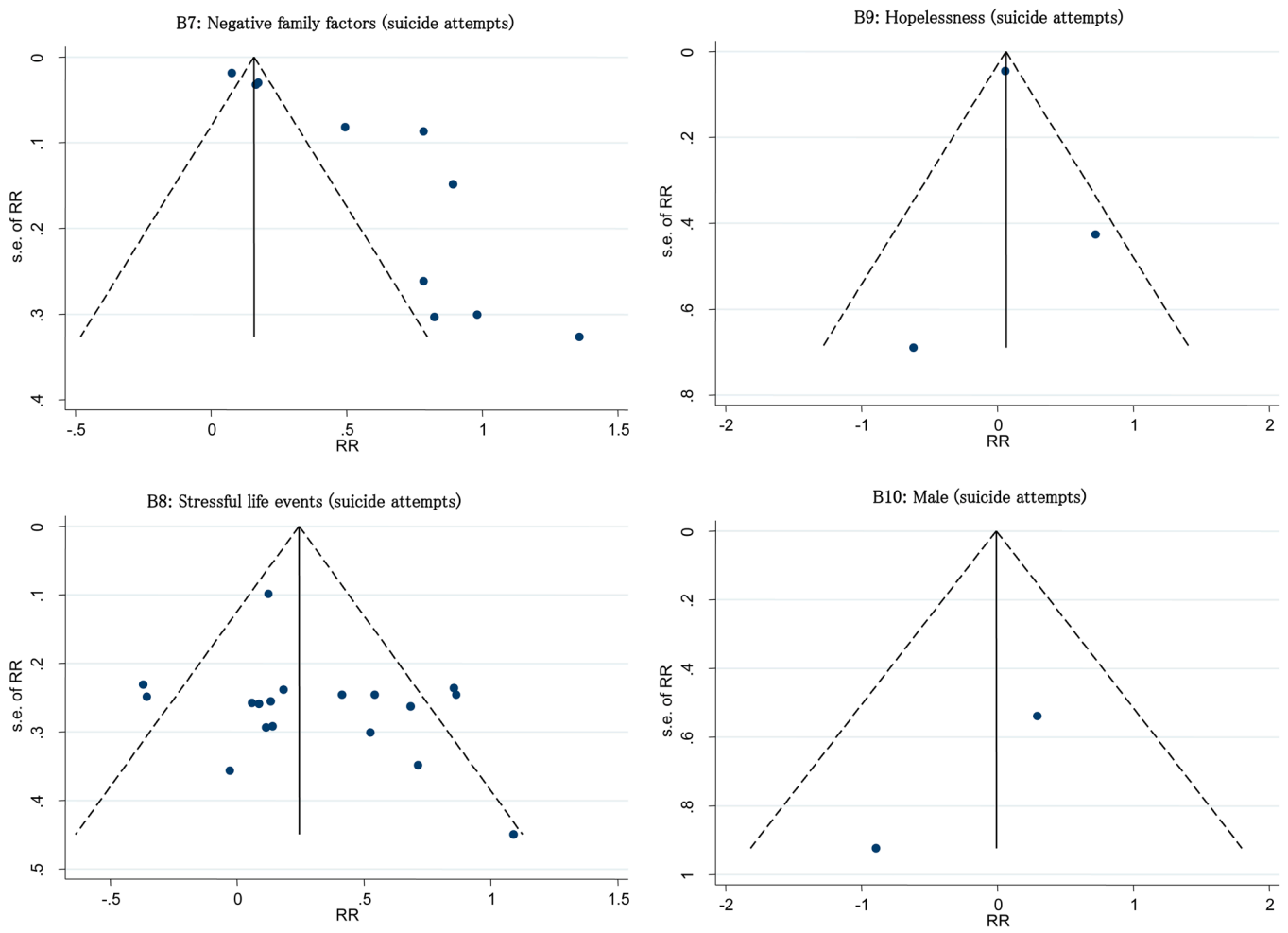


Fig. 3. (*continued*).

symptoms and disorders for suicide and suicide attempts. The publication bias may affect the reliability of the results.

According to Egger's test, there was publication bias associated with negative family factors for suicidal ideation ($p = 0.002$) and suicide attempts ($p = 0.001$) (Table 2). Egger's test found no publication bias associated with predictors of severe depression ($p = 0.247$) and stressful life events ($p = 0.122$) for suicidal ideation, predictors of anxiety symptoms ($p = 0.438$) and stressful life events ($p = 0.225$) for suicide attempts, predictor of anxiety symptoms ($p = 0.773$) for suicide death (Table 2).

4. Discussion

4.1. Main findings

This study presented the predictive factors for suicidal ideation, suicide attempt and suicide death in individuals with MDD. Severe depression, previous suicidal behaviors and alcohol/substance disorders were common risk factors for suicidal ideation, suicide attempt and suicide death. Individuals with MDD who had suffered from negative environmental factors, such as stressful life events, negative school factors and negative family factors, were more likely to have suicidal ideation and attempt suicide. Furthermore, comorbid anxiety was a predictor for suicide attempt and suicide death in MDD individuals.

Severe depression should be seen as a syndrome or a degree of impairment and it was a strong predictor for suicide (Handley et al., 2018). The long-term suicide risk for medium and severe depression was 3.1% and 11.4%, respectively (Brådvik et al., 2008). Severe depression patients have more significant changes in pathology such as hypothalamic-pituitary-adrenal axis changes, structural and functional brain changes and so on, which may contribute to suicide attempt (Malhi and Mann, 2018). People who have previous suicidal behaviors were also at high risk to perform future suicide attempts and committed suicide (Aaltonen et al., 2019; Holma et al., 2010; May et al., 2012; Miranda and Shaffer, 2013; Seo et al., 2015). Individuals' suicidal thoughts or behaviors would get strengthened in MDD patients' memory, reduce the threshold of suicide and then lead to suicide action (Rudd, 2006; Beck, 1996). Besides, cognitive inflexibility, defined as failing to handle changes in the external environmental properly, could explain why people attempting suicide before seemed to be more vulnerable to commit suicide again in the future (Schotte and Clum, 1982, 1987; Miranda et al., 2012).

MDD is strongly associated with environmental factors, such as stressful life events (victims of crime, having trouble with boss or co-workers, interpersonal stresses, etc.) from family context (family history of suicide, childhood sex abuse, etc.) and school context (bad interpersonal affiliations, bad school achievement, etc.) (Wang et al., 2015; Fergusson et al., 2003). As one of the stressful life events, poor interpersonal relationship indicates low social support which is closely associated with suicide related behaviors in MDD individuals (Lewinsohn et al., 1997). Specific genetic factors make some people more vulnerable to stressful life events and more susceptible to depression

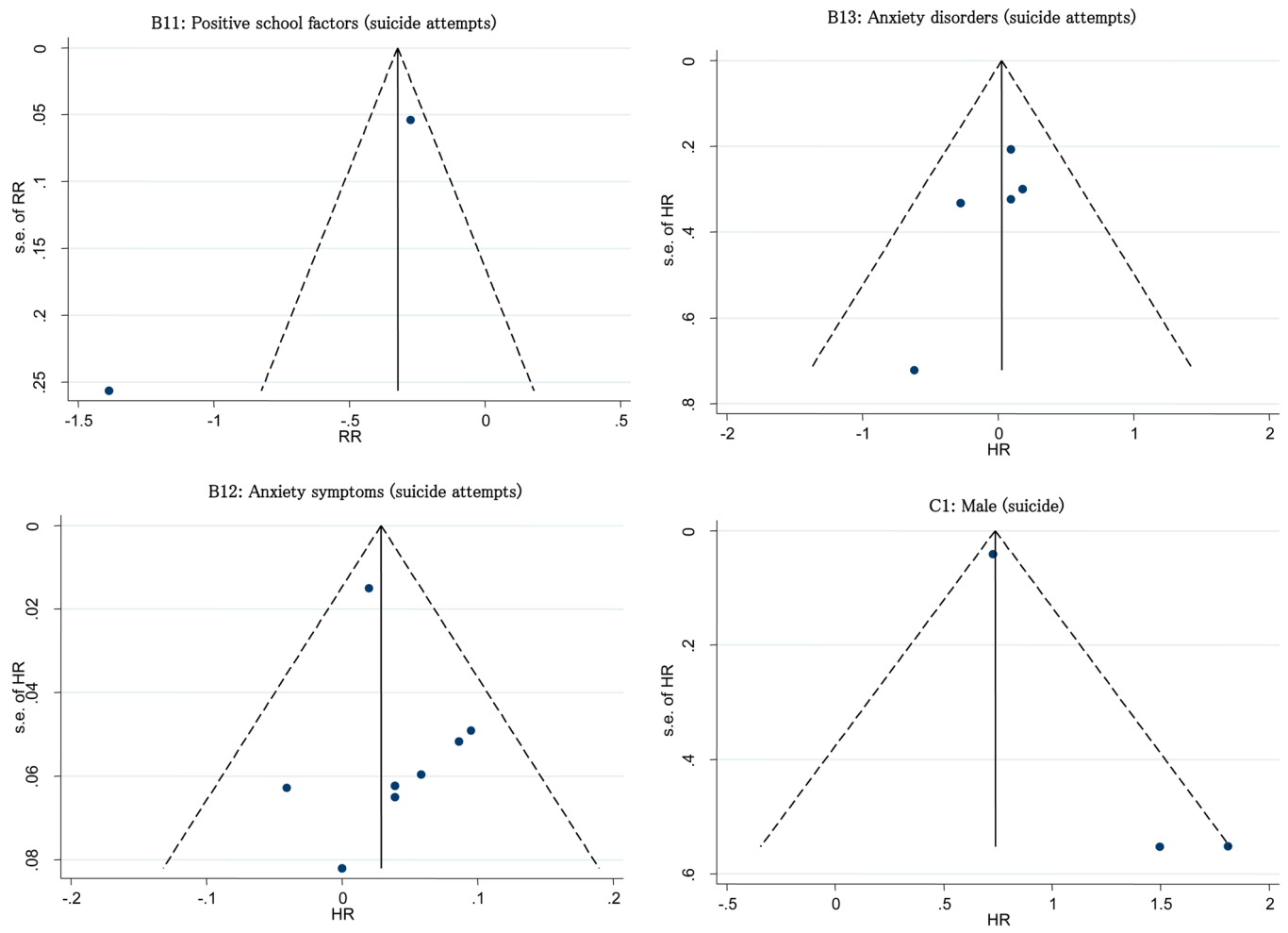


Fig. 3. (*continued*).

(Kendler et al., 1999). Furthermore, MDD patients who have financial stress were prone to suffer from treatment resistant, which might increase the risk of suicidal attempts (Amital et al., 2008; Stegenga et al., 2012). The diversity of individual's reaction to stressful life events and behavior could explain part of the correlations (Wang et al., 2015).

Alcohol use related disorders should be taken into account in the assessment of suicide risk in MDD inpatients, which has important significance for clinical guidance (Hansen et al., 2003). The purpose of drinking is to pursue the pleasant experience. Besides, when people feel bad, drinking may be a kind of adjustment method. However, when an alcohol dependent failed to acquire enough alcohol they need, he/she would resort to make some suicidal related behaviors (Ribeiro et al., 2016). By reducing natural anxiety towards self-mutilation or affecting drug metabolism, alcohol use related disorders could still increase suicide risk in depressed patients characterized by low mood (Boenisch et al., 2010). Anxiety comorbidity could also increase the risk of suicide behaviors. MDD patients have deficits in basic attentional level. On the another hand, comorbid anxiety could cause impairments in higher level executive attention functions among individuals with MDD (Lyche et al., 2011). Anxiety disorders in MDD patients is important for future research in suicide prevention.

Predictors indentified in this study should be considered while developing suicide risk prediction models and risk assessment tools for estimating future suicide in people with MDD (Hawton et al., 2013). The common risk factors for the three outcomes of suicided related behaviors (e.g. previous suicidal behaviors, environmental factors and comorbidity) deserve further research. Environmental factors such as school and family experiences and lifetime grown-up environment context need to be taken into account for the prevention of suicide in clinical practice. Effective school and family supports might play an important role in prevention against future suicide death among people with MDD.

4.2. Limitations

There were several limitations in this study. First, the literature databases were limited, although we had searched different English and Chinese databases regardless of published periods and language restrictions. Second, diverse diagnostic criteria had been used for the definition of suicide related behaviors and MDD. This could lead to high and severe heterogeneity which would impact the reliability of results in this study. Third, there were no enough published studies with high quality for subgroup analysis which should be conducted to explore the potential sources of heterogeneity, such as published year, age of participants, follow-up periods, diagnosis criteria of MDD and diagnosis criteria of suicide related behaviors. Finally, gender played an important role in suicidal related behaviors among people with MDD. However, only two articles for suicide attempts and three articles for suicide death were included on gender differences in this meta-analysis. More researches should be conducted to display the exact correlations between sex and suicide related behaviors among people with MDD.

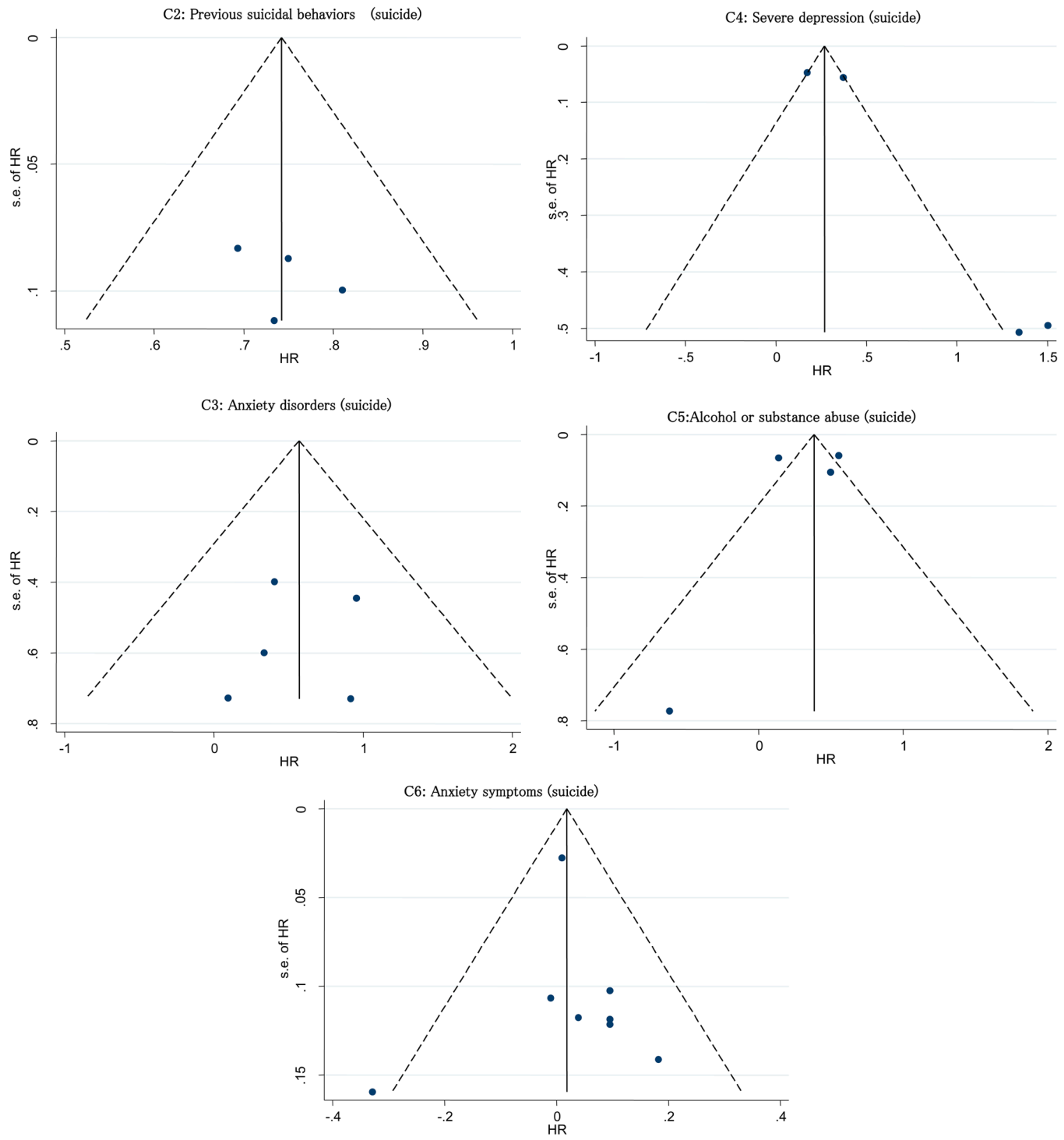


Fig. 3. (*continued*).

5. Conclusions

Previous suicide behaviors, mental disorders (severe depression, anxiety, alcohol/substance disorders) and environmental factors (school and family context, social support, stressful life events, etc.) could contribute to suicide related behaviors in people with MDD. Men suffering from MDD were at higher risk of suicide death than women. The screening and identification of suicide risk individuals among people with MDD would be extremely meaningful for population health. Due to the highly complex course of suicide, it is important to develop theoretical frameworks for suicide interventions among individuals with MDD based on these predictive factors.

Ethical standards

The manuscript does not contain animals, patients, or population data.

Author contributions

YL & JLW contributed to the study design. XL, FM, DL & JZ did literature searches and screening. XL, ML & SY extracted the data. XL &

FM analyzed the data. XL, DL, FM & JZ wrote this manuscript and they contributed equally to this study. YL & JLW were responsible for English revision of the manuscript. All authors reviewed the manuscript.

Declaration of Competing Interest

The authors declared no conflicts of interests.

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